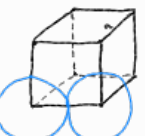


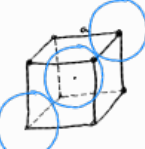
Politično razmerje: $\eta = \frac{V_{\text{atom}}}{V_{\text{režnja}}}$, mrežna razdalja: a

1) (Enostavna) navadna krist. mreža (SC-simple cubic)



$$\eta_{sc} = \frac{8 \cdot \frac{4}{3} \pi r^3}{a^3} = \frac{4 \pi (\frac{a}{2})^3}{3 a^3} = \frac{\pi}{6} = 52 \%$$

2) Telošno centrirana krist. mreža (BCC-body centered cubic)



$$r = \frac{a\sqrt{3}}{4}$$

$$\eta = \frac{(8 \cdot \frac{4}{3} + 1) \frac{4}{3} \pi r^3}{a^3} = \frac{8 \pi \sqrt{3}}{3 \cdot 4^3} = \frac{\pi \sqrt{3}}{6} = 68 \%$$

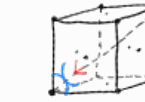
3) Ploskovno centrirana krist. mreža (FCC-face centered cubic)



$$r = \frac{a\sqrt{2}}{4}$$

$$\eta = \frac{(8 \cdot \frac{4}{3} + 6 \cdot \frac{1}{2}) \frac{4}{3} \pi r^3}{a^3} = \frac{16 \pi \sqrt{2}}{3 \cdot 2^3} = \frac{\pi \sqrt{2}}{3} = 74 \%$$

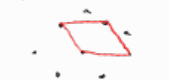
4) Diamantna kristalna mreža
(dva FCC, ena je premaknjena za 1/4 telesne diagonale)



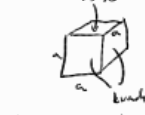
$$r = \frac{\sqrt{3} a}{8}$$

$$\eta = \frac{8 \cdot \frac{4}{3} \pi r^3}{a^3} = \frac{8 \cdot 4 \sqrt{3} \pi}{3 \cdot 8^3} = \frac{\pi \sqrt{3}}{16} = 34 \%$$

5) Trikotna mreža



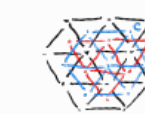
6) Navadna heksagonalna mreža



$$r = \frac{a}{2}$$

$$\eta = \frac{1 \cdot \frac{4}{3} \pi r^3}{\frac{\sqrt{3}}{2} a^2 h} = \frac{\pi}{3\sqrt{3}} = 60 \%$$

7) Najgostejši sklad



ABABAB... → Najgostejši heksagonalni sklad (HCP-hexagonal close-packed)

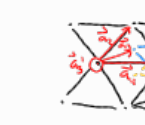
ABCABC... $r = \frac{a}{2}, h = 2 \cdot a \frac{\sqrt{3}}{2}$

$$\eta = \frac{2 \cdot \frac{4}{3} \pi r^3}{\frac{\sqrt{3}}{2} a^2 h} = \frac{\pi}{3\sqrt{2}} = 74 \% = \eta_{fcc}$$

Bravaisova mreža

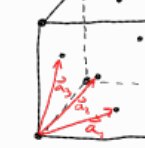
3D: $\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3, n_i \in \mathbb{Z}$

2D: $\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2$



- trikotna mreža (ravnina) lahko opišemo z Brav. mrežo
- hcp (AB)-ABABAB... ni moremo opisati z Brav. mrežo (2 $\vec{a}_3$ pride do plasti C)
- če izberemo $\vec{a}_3$ tako, da knže iz ravnine pravokotno do naslednje plasti A, dobimo Brav. rešetko za polovico delcev →
- $\vec{X} = \vec{R} + \vec{r} \rightarrow \{\vec{r}_1 = 0, \vec{r}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{1}{3})\}$; $\vec{r}$ -baza
- $4 n_1 \vec{a}_1 + n_2 \vec{a}_2 = n_3 \vec{a}_3$
- hcp je navadna heks. mreža z bazo
- ABC mrežo lahko opišemo z Brav. mrežo ($\vec{a}_1, \vec{a}_2, \vec{a}_3$)

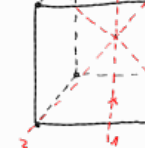
→ fcc



- fcc je Brav. mreža
- $|\vec{a}_1| = |\vec{a}_2| = |\vec{a}_3| = \frac{a}{\sqrt{2}}$
- $\vec{a}_1 = \frac{a}{2}(1, 1, 0)$
- $\vec{a}_2 = \frac{a}{2}(1, 0, 1)$
- $\vec{a}_3 = \frac{a}{2}(0, 1, 1)$
- $\cos \varphi_{12} = \frac{\vec{a}_1 \cdot \vec{a}_2}{|\vec{a}_1| |\vec{a}_2|} = \frac{(\frac{a}{2})^2}{\frac{a}{\sqrt{2}} \cdot \frac{a}{\sqrt{2}}} = \frac{1}{2}$
- $\varphi_{ij} = 60^\circ$
- ⇒ **fcc = ABC mreža**

→ sc, bcc, fcc

Katero točkorno simetrije (vsaj 1 točka ostane na istem mestu) lahko opravimo na sc, bcc, fcc?



1: rotacija za 90° 4-števna os ($4 \times 90^\circ = 360^\circ$); $C_4^k, k \in \mathbb{Z}$; možna operacija: C_4^1, C_4^2, C_4^3 ; $C_4^4 = C_4^0 = E$ (identiteta)

→ lahko že 2 podobni osi → 8 netrivialnih operacij in 1 netrivialna

2: rotacija za 120° 3-števna os $C_3^k, C_3^2, C_3^3 = E$, 4 možne → 8 netrivialnih

3: rotacija za 180° 2-števna os $C_2^k, C_2^2 = E$; 6 možnih → 6 netrivialnih

} skupno 24 operacij

4: inverzija $\vec{R} \rightarrow -\vec{R}$ $I \rightarrow -1$

Razlika med rotacijo in inverzijo:

operacijo zapišemo z matriko

• $\det(\text{matrica}) = 1$ ortogonalna
• $\det(\text{inverzija}) = -1$ $\vec{R} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \vec{R}$

→ če kombiniramo inverzijo z rotacijami, dobimo skupno $24 \cdot 2 = 48$ operacij

Pokaži, da te operacije tvorijo točkasto simetrijsko grupo.

$g \in G$

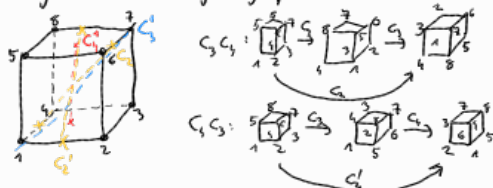
• $\exists g \in V$

• $\forall g \exists g^{-1} \in V$

• $\forall g_1, g_2 \rightarrow g_1 g_2 \in G$

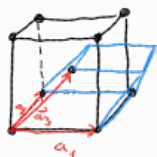
• $g_1(g_2 g_3) = (g_1 g_2) g_3$

→ pokazati smo za 1 primer, načeloma bi mogli pokazati za vse



→ ker je vrstni red operacij pomenben, je to nekomutativna grupa

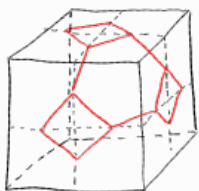
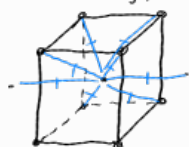
Wigner-Seitzova osh. celica



$V = \frac{a^3}{2}$, paralelepiped je primitivna osh. celica, ampak ruši simetrijo

W-S. osh. celica ohranja prej pokazane simetrije

• dobimo jo, če središčno kroglico povežemo s sosedi in drugimi sredi, nato po daljico prepolovimo. Ravnine sestavljajo W-S celico



Ortorombске Brav. mreže

$a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$



enostavna



telesno centrirana



ploskovno centrirana



base-centered orthorombic

Tetragonalne BM

$a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$



enaka struktura enaka struktura
kot telesno centrirana mreža kot enostavna mreža

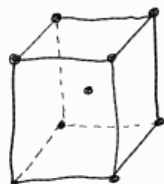
Recipročna mreža bcc

$\vec{R} \dots$ Brav. mreža

$\vec{K} : e^{i\vec{K} \cdot \vec{R}} = 1 \forall \vec{R}$

$\vec{K} = m_1 \vec{A}_1 + m_2 \vec{A}_2 + m_3 \vec{A}_3$, $m_i \in \mathbb{Z}$

$\vec{A}_1 = \frac{\vec{a}_2 \times \vec{a}_3}{(\vec{a}_1, \vec{a}_2, \vec{a}_3)} 2\pi$



$\vec{a}_1 = \frac{a}{2} (-1, 1, 1)$

$\vec{a}_2 = \frac{a}{2} (1, -1, 1)$

$\vec{a}_3 = \frac{a}{2} (1, 1, -1)$

$\vec{a}_2 \times \vec{a}_3 = \frac{a^2}{4} (0, 2, 2) = \frac{a^2}{2} (0, 1, 1)$

$\vec{a}_3 \times \vec{a}_1 = \frac{a^2}{4} (2, 0, 2) = \frac{a^2}{2} (1, 0, 1)$

$\vec{a}_1 \times \vec{a}_2 = \frac{a^2}{2} (1, 1, 0)$

$(\vec{a}_1, \vec{a}_2, \vec{a}_3) = \frac{a^3}{4} (0, 1, 1) \cdot (-1, 1, 1) = \frac{a^3}{2}$

$\vec{A}_1 = \frac{2\pi}{a} (0, 1, 1)$

$\vec{A}_2 = \frac{2\pi}{a} (1, 0, 1)$

$\vec{A}_3 = \frac{2\pi}{a} (1, 1, 0)$

Preizkus: $\vec{a}_i \cdot \vec{A}_j = 2\pi \delta_{ij}$

*rec. mreža je ploskovno centrirana

rec. mreža trikotne mreže

$$\begin{pmatrix} \vec{a}_1 \\ \vec{a}_2 \end{pmatrix} (\vec{a}_1, \vec{a}_2) = 2\pi I \Rightarrow (\vec{a}_1, \vec{a}_2) = 2\pi \begin{pmatrix} \vec{a}_1 \\ \vec{a}_2 \end{pmatrix}^{-1} = 2\pi \frac{1}{a} \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}^{-1} =$$

$$\vec{a}_1 = a(1, 0) = \frac{2\pi}{a} \frac{2}{\sqrt{3}} \begin{bmatrix} \frac{\sqrt{3}}{2} & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$\vec{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$$

$$\rightarrow \vec{A}_1 = \frac{4\pi}{\sqrt{3}a} (\frac{\sqrt{3}}{2}, -\frac{1}{2})$$

$$\vec{A}_2 = \frac{2\pi}{\sqrt{3}a} (0, 1)$$



Opis fcc

1) fcc

2) sc+baza (konvencionalna osh. celica)

3) uvedena heksagonalna baza

$$1) \vec{a}_1 = a(\frac{1}{2}, \frac{1}{2}, 0)$$

$$2) \vec{a}_1 = a(1, 0, 0)$$

$$3) \vec{a}_1 = \frac{a}{\sqrt{2}}(1, 0, 0)$$

$$\vec{a}_2 = a(\frac{1}{2}, 0, \frac{1}{2})$$

$$\vec{a}_2 = a(0, 1, 0)$$

$$\vec{a}_2 = \frac{a}{\sqrt{2}}(\frac{1}{2}, \frac{1}{2}, 0)$$

$$\vec{a}_3 = a(0, \frac{1}{2}, \frac{1}{2})$$

$$\vec{a}_3 = a(0, 0, 1)$$

$$\vec{a}_3 = \frac{a}{\sqrt{2}}(0, 0, 3) = a\sqrt{3}(0, 0, 1)$$

$$\vec{r}_1 = 0$$

$$\vec{r}_1 = 0$$

$$\vec{r}_1 = 0$$

$$\vec{R} = n_1\vec{a}_1 + n_2\vec{a}_2 + n_3\vec{a}_3 + \vec{r}_1$$

$$\vec{R} = \frac{a}{2}(1, 1, 0)$$

$$\vec{R} = \frac{a}{3}(\vec{a}_1 + \vec{a}_2 + \vec{a}_3)$$

$$\vec{r}_2 = \frac{a}{2}(1, 0, 1)$$

$$\vec{r}_2 = \frac{a}{3}(\vec{a}_1 + \vec{a}_2 + \vec{a}_3)$$

$$\vec{R} = n_1\vec{a}_1 + n_2\vec{a}_2 + n_3\vec{a}_3 + \vec{r}_1$$

Geometrijski strukturni faktor

$$S_{\vec{k}} = \sum_i e^{-i\vec{k} \cdot \vec{r}_i}$$

$$1) \vec{R} = m_1\vec{A}_1 + m_2\vec{A}_2 + m_3\vec{A}_3 = \frac{2\pi}{a}(m_1, m_2, m_3)$$

$$S_{\vec{k}} = 1$$

$$\vec{A}_1 = \frac{2\pi}{a}(1, 0, 0)$$

$$\vec{A}_2 = \frac{2\pi}{a}(0, 1, 0)$$

$$\vec{A}_3 = \frac{2\pi}{a}(0, 0, 1)$$

$$S_{\vec{k}} = 1 + e^{-i\pi(m_1+m_2)} + e^{-i\pi(m_1+m_3)} + e^{-i\pi(m_2+m_3)}$$

$$m_i \text{ sodni: } S_{\vec{k}} = 4$$

$$m_i \text{ lihi: } S_{\vec{k}} = 0$$

$$2 \text{ sodni, 1 lihi: } S_{\vec{k}} = 0$$

$$2 \text{ lihi, 1 sodni: } S_{\vec{k}} = 0$$

$$\frac{S_{\vec{k}}}{4} = \frac{1}{4}$$

$$\frac{S_{\vec{k}}}{4} = \frac{3}{4}$$

$$\frac{S_{\vec{k}}}{4} = \frac{1}{4}$$

$$\frac{S_{\vec{k}}}{4} = \frac{3}{4}$$

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$$\frac{S_{\vec{k}}}{4} = \frac{1}{4}$$

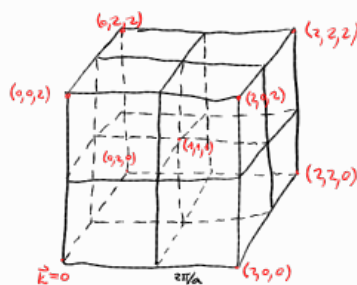
$$\frac{S_{\vec{k}}}{4} = \frac{3}{4}$$

$$\frac{S_{\vec{k}}}{4} = \frac{1}{4}$$

$$\frac{S_{\vec{k}}}{4} = \frac{3}{4}$$

$$\frac{S_{\vec{k}}}{4} = \frac{1}{4}$$

Rec. mreža



• z = označene točke, ki dajejo nenuljni \$S_{\vec{k}}\$

• dobili smo bcc \$\rightarrow\$ \$S_{\vec{k}}\$ popravi obratno fcc kot sc+baza

$$3) \vec{R} = m_1\vec{A}_1 + m_2\vec{A}_2 + m_3\vec{A}_3$$

$$S_{\vec{k}} = 1 + e^{i\pi(m_1\vec{A}_1 + m_2\vec{A}_2 + m_3\vec{A}_3) \cdot \frac{1}{3}(\vec{a}_1 + \vec{a}_2 + \vec{a}_3)}$$

$$+ e^{\frac{2\pi i}{3}(m_1\vec{A}_1 + m_2\vec{A}_2 + m_3\vec{A}_3) \cdot (\vec{a}_1 + \vec{a}_2 + \vec{a}_3)} =$$

$$= 1 + e^{\frac{2\pi i}{3}(m_1 + m_2 + m_3)} + e^{\frac{4\pi i}{3}(m_1 + m_2 + m_3)}$$

$$\Sigma m_i = 3N: S_{\vec{k}} = 3$$

$$\Sigma m_i = 3N+1: S_{\vec{k}} = 1 + e^{\frac{2\pi i}{3}} + e^{\frac{4\pi i}{3}} = 0$$

$$\Sigma m_i = 3N+2: S_{\vec{k}} = 1 + e^{\frac{4\pi i}{3}} + e^{\frac{2\pi i}{3}} = 0$$

$$\lambda = 2\text{\AA}$$

1) praznina vzorec

$$\Theta_1 = 83,6^\circ$$

$$\Theta_2 = 11,1^\circ$$

2) monokristal

$$\Theta = 11,1^\circ$$

$$\frac{I}{I_0} = \frac{1}{4}$$

(pri različnih prost. kotih dobimo razl. intenziteto)



$$\vec{a}_1 = a(1, 0, 0)$$

$$\vec{a}_2 = a(0, 1, 0)$$

$$\vec{a}_3 = a(0, 0, 1)$$

$$\vec{r}_1 = 0$$

$$\vec{r}_2 = c(1, 1, 1)$$

$$c = ?$$

1)



$$k - k' = \frac{2\pi}{\lambda}$$

$$2d \sin \frac{\Theta}{2} = n\lambda \rightarrow 2d \sin \frac{\Theta}{2} = \lambda$$

$$\vec{k} \text{ najkrajši}$$

$$\vec{k} \text{ poljuh}$$

$$\text{izbrani smeri}$$

$$\text{upr. } \vec{k} = \frac{2\pi}{a}(2, 0)$$

$$\text{ampak } \vec{k} = \frac{2\pi}{a}(1, 0)$$

$$S_{\vec{k}} = 1 + e^{-i\frac{2\pi}{a}(m_1 + m_2 + m_3)}$$

$$d = \frac{2\pi}{|\vec{k}|} = \frac{1}{\sqrt{m_1^2 + m_2^2 + m_3^2}}$$

$$\sin \frac{\Theta}{2} = \frac{\lambda}{2a} \sqrt{m_1^2 + m_2^2 + m_3^2} < \frac{\pi}{2}$$

c razdalja do naslednjega atoma

$$\Theta_1 \rightarrow (100), (010), (001) \rightarrow \{100\}$$

$$\Theta_2 \rightarrow \{110\} \rightarrow (110)(101)(011)$$

$$(1\bar{1}0)(10\bar{1})(01\bar{1})$$

$$\uparrow$$

$$\uparrow$$

$$\Theta_1 \rightarrow \{110\}$$

$$\Theta_2 \rightarrow \{200\}$$

$$(bcc)$$

$$I = |S_{\vec{k}}|^2 F(\Theta)$$

bcc: vse ravnine - (200), (020), (002) imajo enak strukturni faktor

\$\Rightarrow\$ intenziteta bi morala biti konstantna

\$\Rightarrow\$ hi bcc

$$\Theta_1 \rightarrow \{110\} \rightarrow (110), (101), (011), (1\bar{1}0), (10\bar{1}), (01\bar{1})$$

$$S_{\vec{k}} = 1 + e^{-i\frac{2\pi}{a}(m_1 + m_2 + m_3)}$$

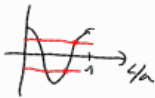
$$S_{\vec{k}} = 2$$

$$S_{\vec{k}} = 2$$

$$\rightarrow \frac{I_2}{I_2'} = \frac{|1 + e^{-i\pi \frac{z}{a}}|^2}{4} = \frac{1 + e^{i\pi \frac{z}{a}} + e^{-i\pi \frac{z}{a}} + 1}{4} = \frac{2(1 + \cos(\pi \frac{z}{a}))}{4} = \frac{4 \cos^2 \frac{\pi z}{2a}}{4} = \frac{1}{4}$$

↓
podatek

$$\cos 2\pi \frac{z}{a} = \pm \frac{1}{2}$$

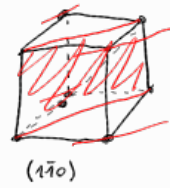
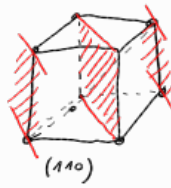


$$2\pi \frac{z_1}{a} = \frac{\pi}{3} \rightarrow c_1 = \frac{a}{6}$$

$$2\pi \frac{z_2}{a} = \frac{2\pi}{3} \rightarrow c_2 = \frac{a}{3}$$

$$2\pi \frac{z_3}{a} = \frac{4\pi}{3} \rightarrow c_3 = \frac{2a}{3}$$

$$2\pi \frac{z_4}{a} = \frac{5\pi}{3} \rightarrow c_4 = \frac{5a}{6}$$



• večja intenziteta,
ker dodatni atom
leži v ravni